$$\sum_j m_{kj}\alpha_j = \beta_k. \tag{3.3}$$

We can write this in matrix form as well. Eq. 3.3 becomes

$$\begin{pmatrix} m_{11} & m_{12} & m_{13} \\ m_{21} & m_{22} & m_{23} \\ m_{31} & m_{32} & m_{33} \end{pmatrix} \begin{pmatrix} \alpha_1 \\ \alpha_2 \\ \alpha_3 \end{pmatrix} = \begin{pmatrix} \beta_1 \\ \beta_2 \\ \beta_3 \end{pmatrix}. \tag{3.4}$$

You're probably familiar with the rule for matrix multiplication, but I will remind you just in case. To compute the first entry on the right, β_1, take the first row of the matrix and "dot" it into the α column:

$$\beta_1 = m_{11}\alpha_1 + m_{12}\alpha_2 + m_{13}\alpha_3.$$

For the second entry, dot the second row of the matrix with the α column:

$$\beta_2 = m_{21}\alpha_1 + m_{22}\alpha_2 + m_{23}\alpha_3.$$

And so on. If you are not familiar with matrix multiplication, run to your computer and look it up right away. It's a crucial part of our tool kit, and I will assume you know it from now on.

There are both advantages and disadvantages to representing vectors and linear operators concretely with columns, rows, and matrices (known collectively as *components*). The advantages are obvious. Components provide a completely explicit set of arithmetic rules for working the machine. The disadvantage is that they depend on a specific choice of basis vectors. The underlying relationships between vectors and operators is independent of the particular basis we choose, and the concrete representation obscures that fact.

3.1.2 Eigenvalues and Eigenvectors